Yaswanth Muppalla

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Senior Software Engineer | ***Java/Python Backend*** | ***Distributed Systems & APIs*** | ***AWS/GCP*** | ***AI Platforms***

PROFESSIONAL EXPERIENCE

Senior Software Engineer | American Express — Apr 2023 - Present

Impact: Built a **cross-cloud GCP-to-AWS data and API platform** processing **10M+ financial records**, improving reporting efficiency by **35%**, achieving **99% data accuracy**, and reducing rerun/debug cycles by **3x**.

- Built **JSON-configurable GCP ingestion pipelines** for **FIS files**, translating BCA/CCA reporting and reconciliation rules into reusable processing components and **AI-ready data contracts** for governed retrieval, reporting, and **downstream copilot APIs**.
- Implemented **Kafka-backed ingestion**, **Apache Flink processing**, and **Airflow orchestration** for **validation, joins, deduplication, enrichment**, dependency controls, reruns, and **failure recovery** across reporting and **AI workloads** with **audit-ready execution**.
- Published governed outputs to **BigQuery curated tables** with **quality checks, lineage, and audit controls** for consumption by APIs, reporting applications, and **downstream AI copilots** through **stable service contracts** and traceable access patterns.
- Exposed curated data through **AWS API Gateway and Lambda services** for **analytical reads and low-latency reporting**, with **API contracts, retries, SLA monitoring, and tracing** for reporting and **copilot consumers**.
- Designed an **IRIS-backed AI ingestion and semantic retrieval layer** with document extraction, chunking, metadata enrichment, scoped search, and batch publishing for **grounded copilot retrieval** with source-level traceability and **retrieval quality controls**.
- Developed **AI copilot services** with **agent/context orchestration**, source-ledger tracking, controlled prompt assembly, **citations, guardrails**, **evaluation traces**, and audit-friendly response logs for production support and **response-quality diagnostics**.

Software Engineer | Skechers — Oct 2020 - Mar 2023

Impact: Delivered **15+ Python APIs and event-driven services**, maintained **99.9% uptime**, reduced infrastructure costs by **22%**, migrated **35M+ records**, and cut incident resolution time by **40%**.

- Developed **15+ Python/Pydantic REST APIs** on **AWS Lambda and API Gateway** for loyalty and campaign workflows, sustaining **99.9% uptime**.
- Orchestrated **event-driven processing** with **Step Functions**, retries, timeout handling, and downstream service calls, **isolating failures** across distributed loyalty operations.
- Applied **layered service architecture** across **validation, orchestration, data access**, retry handling, and downstream integrations.
- Migrated **35M+ SFMC records** to **MongoDB** and improved query performance by **50%** through indexing and caching.
- Integrated **SessionM, SFMC, MongoDB, and AWS services** for loyalty workflows with **sub-500ms transactions**.
- Improved **production resilience** with **CloudWatch/Datadog alerts**, **API failure tracing**, retries, and SLA-focused incident response.

Full-Stack Developer | SOCH Inc. - Richmond, VA — Jan 2020 - Sep 2020

- Built a **Spring Boot ticketing platform** with **REST microservices**, validation, exception handling, and **service-layer boundaries** for government workflows.
- Designed **Spring REST services** with API controllers, **request validation, exception handling**, and **service-layer separation** for maintainable government workflows.
- Used **Java ExecutorService/CompletableFuture** for **asynchronous** ticket updates, notifications, and background job execution.
- Managed **concurrency** with **thread-safe collections**, **synchronization**, timeouts, and full-stack operational workflow screens.

FOUNDER-LED 0-TO-1 TECHNICAL PROJECT

OmInsights - Founder-Led Backend & AI Platform Build — **Jan 2024 - Present**

Impact: Built a **six-component backend and AI platform** with **tenant-scoped APIs**, async processing, **CDK-managed AWS infrastructure**, **ECS/Fargate services**, and controlled agent workflows with **evaluation and traceability** across tenant-aware AI responses.

- Architected **FastAPI services** with **auth/tenant isolation**, typed REST contracts, **repository/service boundaries**, **MongoDB/OpenSearch** persistence, and background workers for **AI-enabled application workflows** and **policy-aware orchestration**.
- Provisioned **AWS environments with CDK and CI/CD**, deploying containerized Context Engine/MCP services to **ECS/Fargate** with **S3/SQS**, secrets/config, autoscaling, and observability for **backend and agent services** across repeatable environments.
- Implemented **asynchronous document processing** through **S3, SQS router/worker Lambdas**, **DLQs, idempotency keys**, and MongoDB/OpenSearch publishing with **retry-aware error classification**, reliable reprocessing, and **retrieval indexing**.
- Built **MCP/API runtime patterns** and a **shared LLM gateway** for model routing, tool execution, **retries/fallbacks**, usage logging, and execution traces for **production agent workflows**, model/provider behavior, and **operational diagnostics**.
- Engineered **context orchestration and retrieval** with session state, selected documents, source ledgers, **LLM-as-a-judge evaluations**, **guardrails, and citations** for **grounded, policy-aware responses** under **tenant-scoped context controls**.

TECHNICAL SKILLS

Languages & Frameworks: Java 17, **Python**, Scala, SQL, **Spring Boot**, **FastAPI**, Pydantic, TypeScript/Angular
Backend & Distributed Systems: REST APIs, **microservices**, **event-driven architecture**, service contracts, **Java concurrency**, async workers, caching, retries, idempotency
Cloud & DevOps: AWS, **GCP**, Lambda, API Gateway, **ECS/Fargate**, Step Functions, S3/SQS, Docker, **AWS CDK/CloudFormation**, CI/CD
Data & Streaming: Kafka, **Flink**, **Airflow**, **BigQuery**, MemSQL, MongoDB, OpenSearch/Elasticsearch, batch/stream processing
Observability: Datadog, **CloudWatch**, Kibana, X-Ray, ELK Stack, **structured logging**, SLA alerts, failure tracing
AI Platform Systems: agent/context orchestration, **MCP/FastMCP**, **LLM gateways**, RAG/vector search, LLM-as-a-judge evaluations, guardrails, citations

EDUCATION

State University of New York, New Paltz | Master of Science, Computer Science | CGPA: 3.6/4.0 — May 2020

JNTUK | Bachelor of Technology | CGPA: 8.5/10 — 2018